# Enhanced TCP Petra: Early-Exit Slow Start for Buffer-Aware Congestion Control

Project Proposal

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# Outline

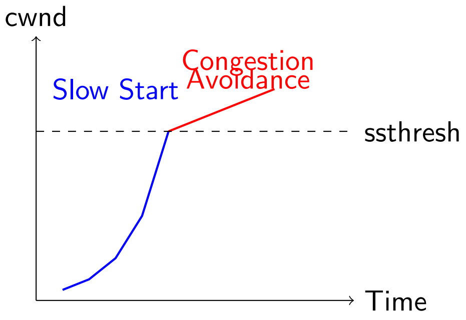
# TCP Congestion Control

**Key Phases:**

- **Slow Start**: Exponential cwnd growth
- **Congestion Avoidance**: Linear growth
- **Fast Recovery**: Handle packet loss

**Challenge:**

- Finding optimal ssthresh value
- Avoiding bufferbloat
- Balancing throughput vs. latency



# TCP Westwood & Petra

**TCP Westwood :**

- End-to-End bandwidth estimation
- Monitors ACK rate at sender
- Sets ssthresh based on estimated bandwidth
- Formula: $ssthresh = ELC \times MinRTT$

**Petra Enhancement :**

- Improved bandwidth estimation using ACK counting
- **Faster Start**: Accelerated cwnd growth in slow start
- Better performance in wireless networks
- Still suffers from bufferbloat in small buffer scenarios

# The Problem: Bufferbloat During Slow Start

## Problem Description

Petra's "Faster Start" aggressively increases cwnd during slow start phase. When bottleneck buffers are small, this causes:

- **RTT Spike**: Queue buildup increases latency
- **Bufferbloat**: Excessive queueing delay
- **Poor User Experience**: High latency for interactive applications

**Observed Behavior:**

- Cwnd overshoots optimal value
- RTT increases 2-3x during slow start
- Packet drops trigger slow start reset

**Impact:**

- × High latency
- × Jitter in real-time apps
- ✓ High throughput achieved

# Our Solution: Early-Exit Slow Start

## Core Idea

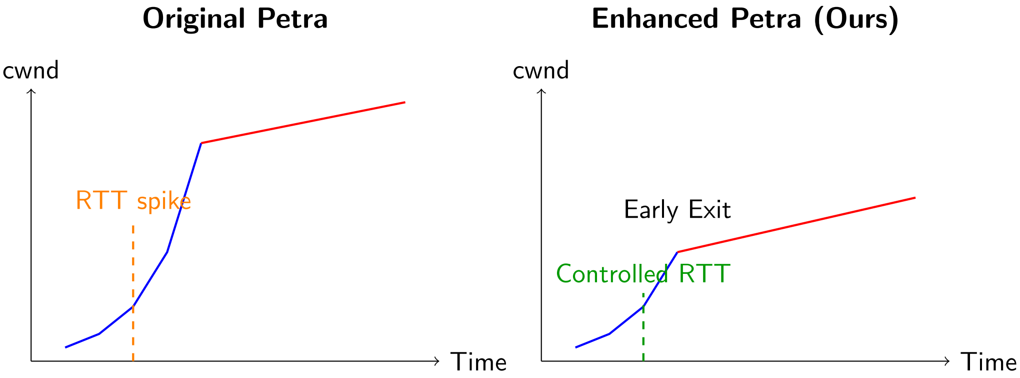
Monitor RTT during slow start. If RTT increases significantly above baseline (minRTT), exit slow start early to prevent bufferbloat.

**Key Components:**

1. **RTT Monitoring**: Track minimum RTT (minRTT) and current RTT
2. **Queue-Delay Threshold**: Define acceptable RTT increase (e.g., 20%)
3. **Early Exit Trigger**: Force transition to congestion avoidance when threshold exceeded

**Condition:** If $currentRTT > minRTT \times 1.20$
**Action:** $cwnd \leftarrow ssthresh$, enter Congestion Avoidance

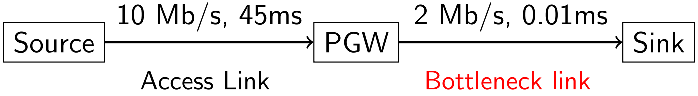
# Visual Comparison




- Blue: Slow Start phase
- Red: Congestion Avoidance phase
- Orange/Green dashed: RTT behavior

# Simulation Setup

**Network Topology (from paper):**




**Simulation Parameters:**

- Access link: 10 Mbps, 45ms delay
- Bottleneck link: 2 Mbps, 0.01ms delay
- Error model: Uniform Error Model (PER: 0.005)
- Application: Bulk Send Application
- Simulation time: 5-10 seconds

# Performance Metrics

**Evaluation Metrics:**

1. **Throughput**: Total data transmitted over time
2. **Congestion Window (cwnd)**: Growth pattern and stability
3. **Round-Trip Time (RTT)**: Latency and queue delay
4. **Packet Loss Rate**: Network efficiency

**Comparison Protocols:**

- TCP Westwood (baseline)
- TCP NewReno (baseline)
- Petra (original)
- **Enhanced Petra** (our modification)

**Experimental Variables:**

- Bottleneck bandwidth: 2-10 Mbps
- Propagation delay: 1-60 ms
- Packet error rate: 0.005-0.05

# Potential Challenges

## Challenge : Parameter Tuning

**Issue:** Finding optimal RTT threshold (20%?)

**Mitigation:** Run sensitivity analysis with multiple threshold values (10%, 15%, 20%, 25%)

# Thank You!